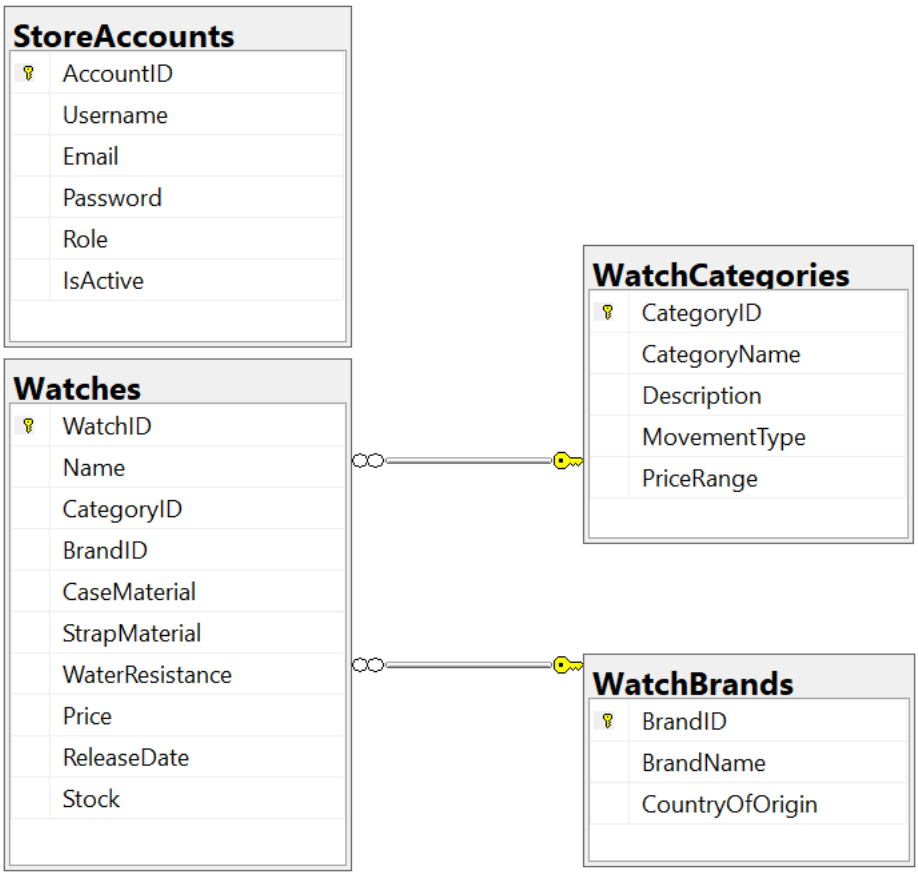
Assignment 02 for MSS301 - Watch Management using Saga Pattern

1. Introduction

In this Assignment, you will build a microservices-based system for a watch store. The primary challenge is to maintain data consistency across different services without using traditional distributed transactions (like two-phase commit), which are often unsuitable for a microservices architecture.



- **Projects in IntelliJ IDEA:**
    1. MSAccount_StudentCode: Manages user accounts (StoreAccounts table).
    2. MSBrand_StudentCode: Manages watch brands (WatchBrands table).

3. MSWatch_StudentCode: Manages watch products and categories (Watches, WatchCategories tables).
4. APIGateway_StudentCode: Acts as the single entry point for all client requests, routing them to the appropriate microservice.

- **MS SQL Databases:**
  - You will use JPA/Hibernate with a **code-first approach**. This means you will define your data models as Java @Entity classes, and Hibernate will automatically generate the database schema (tables, columns, relationships) for you upon application startup.

1. DBAccount: Will contain the StoreAccounts table.
2. DBBrand: Will contain the WatchBrands table.
3. DBWatch: Will contain the Watches and WatchCategories tables.

Each microservice will connect **only** to its own dedicated database, ensuring true service independence.

Table data type

StoreAccounts

Column Name	Data Type	Description
AccountID	INT (Primary Key)	Unique identifier for each system account.
Username	VARCHAR(20) NOT NULL	Login username for the account.
Email	VARCHAR(100) NOT NULL	Email address associated with the account.
Password	VARCHAR(100) NOT NULL	The password associated with the account.
Role	int	User role (e.g., Administrator (1), Staff (2), Member (3)).
IsActive	BOOLEAN DEFAULT TRUE	Indicates if the account is active (TRUE/FALSE).

WatchCategories

Column Name	Data Type	Description

CategoryID	INT (Primary Key)	Unique identifier for each watch category.
CategoryName	VARCHAR(100)	Name of the watch category.
Description	TEXT	Short description of the category.
MovementType	VARCHAR(50)	The movement type (e.g Automatic or Quartz or Manual)
PriceRange	VARCHAR(100)	Typical price range (e.g., $10-$50, $50-$100).

Watches

Column Name	Data Type	Description
WatchID	INT (Primary Key)	Unique identifier for each watch.
Name	VARCHAR(255)	Name of the watch.
CategoryID	INT (Foreign Key)	Links to WatchCategories(CategoryID).
BrandID	INT (Foreign Key)	Links to WatchBrand(BrandID).
CaseMaterial	VARCHAR(100)	Case material.
StrapMaterial	VARCHAR(100)	Strap material.
WaterpMaterial	VARCHAR(50)	Water material.
Price	DECIMAL(10,2)	Price of the watch.
ReleaseDate	DATE	Date when the watch was released.
Stock	INT	Number of units available in stock.

WatchBrand

Column Name	Data Type	Description
BrandID	INT (Primary Key)	A unique identifier for each brand.
BrandName	VARCHAR(200)	The name of the brand that produces the watch.
CountryOfOrigin	VARCHAR(50)	The country where the brand originated.

2. Data *(must input these information)*

StoreAccounts

AccountID	Username	Email	Password	Role	IsActive
1	adminsys	admin@drawingpen.com.vn	@1	1	1
2	staff02	staff02@drawingpen.com.vn	@1	2	1
3	member01	member01@drawingpen.com.vn	@1	3	1
4	member02	member02@drawingpen.com.vn	@1	3	0
5	staff01	staff01@drawingpen.com.vn	@1	2	0

WatchCategories

CategoryID	CategoryName	Description	Movement Type	PriceRange
1	Dive Watches	Designed for underwater diving and feature high water resistance.	Automatic	$500 - $10,000
2	Chronograph Watches	Watches with a stopwatch function in addition to the main display.	Automatic or Quartz	$300 - $8,000
3	Dress Watches	Elegant and simple, intended to be worn with formal wear.	Automatic or Manual	$400 - $20,000
4	Digital Watches	Watches with an LCD or LED screen that show time in a digital format.	Quartz	$50 - $500
5	Pilot Watches	Originally designed for aviators, featuring large, easy-to-read dials.	Automatic	$700 - $15,000

WatchBrands

BrandID	BrandName	CountryOfOrigin
1	Rolex	Switzerland
2	Omega	Switzerland
3	Seiko	Japan
4	Casio	Japan
5	TAG Heuer	Switzerland

Watches

Wat	Name	Categ	Bra	Case	StrapM	WaterR	Pric	Relea	St

chID		oryID	ndID	Material	aterial	esistance	e	seDate	ock
1	Submariner Date	1	1	Oyster steel	Oysterst eel	300 meters	9550.00	2020-09-01	50
2	Speedmaster Moonwatch Professional	2	2	Stainless Steel	Stainless Steel	50 meters	7200.00	2021-01-05	75
3	Presage Cocktail Time	3	3	Stainless Steel	Leather	50 meters	425.00	2019-06-01	200
4	G-Shock DW-5600E-1V	4	4	Resin	Resin	200 meters	69.95	1996-06-12	500
5	Carrera Calibre Heuer 02	2	5	Stainless Steel	Alligator Leather	100 meters	5900.00	2022-03-30	60
6	Seamaster Diver 300M	1	2	Stainless Steel	Rubber	300 meters	5400.00	2018-02-15	120
7	Prospex SPB143	1	3	Stainless Steel	Stainless Steel	200 meters	1200.00	2020-07-01	150
8	Daytona	2	1	Oyster steel	Oysterst eel	100 meters	14550.00	2016-03-16	30
9	F-91W	4	4	Resin	Resin	30 meters	19.99	1989-01-01	1000
10	IWC Pilot's Watch Mark XVIII	5	4	Stainless Steel	Leather	60 meters	4500.00	2016-01-18	80

2. Assignment Objectives

Implement CRUD actions with Spring Boot Microservices using Saga Pattern and Database per Service.

1. List all items in *Watches* table (the information includes all information in this table, CategoryName from *WatchCategories* table). **Note that, No permissions for this function.**

2. Check if login successfully with *Administrator* role, delete the selected item in *Watches* table with the updating other related Microservices.

3. Check if login successfully with *Administrator* role, add new item with the following r*equirements:*
   - *The new item will display at the top of the list.*
   - All fields are required.
   - Value for Price >= 100.
   - Value for Name is greater than 10 characters.
   - Value for ReleaseDate = Current date

And then update other related Microservices.

4. Check if login successfully with *Administrator* role, update the existing item with the r*equirements are same as add function.* And then update other related Microservices.

Design Postman Script components/pages.

5. Create Postman Script for testing
   - API Gateway - A single entry point for all client requests, routing traffic to the appropriate downstream microservices.
   - Microservices
     - AccountService: Manages login process.
     - BrandService: Manages information of brands.
     - Watch Service: Manages watch, which will have a relationship with a brand.